

Stepping Stones Sr. Secondary School

SECTOR – 37

CHANDIGARH

BATCH: CDITY224X01

QP CODE: X01T08.P

JEE MAINS – 2024

Time: 3 Hours

Maximum Marks: 300

Please read the instructions carefully.

INSTRUCTIONS

A: General:

1. Immediately fill in the particulars on this page of the Test Booklet with Blue/Black Ball point pen.
2. Darken the appropriate bubbles with **Blue/Black Ball Point Pen only**.
3. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers and electronic gadgets in any form are not allowed.
4. The answer sheet, a machine-gradable Objective Response Sheet (ORS) is provided separately.
5. Do not Tamper/mutilate the **ORS** or this booklet.
6. No additional sheets will be provided for rough work
7. On completion of this test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**

B: Questions paper format and Marking Scheme:

1. The Test Booklet consists of **75** questions. The maximum marks are 300.
2. There are **3** parts in the question paper A, B, C consisting of **Physics, Chemistry & Mathematics** having 25 questions in each part of equal weightage.
3. For each question in **PART A** you will allotted **4 (four)** marks for correct response. Student will be awarded **–1 Marks** as penalty for each incorrect response.
4. For each question in **PART B**, you will be **awarded 4 marks** if you darken the bubble corresponding to the correct answer and zero mark if no bubbles is darkened. In case of bubbling of incorrect answer there is **No Negative** Marking.

Enrollment No. :

Batch : _____

Name : _____

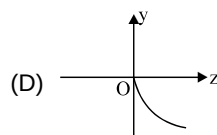
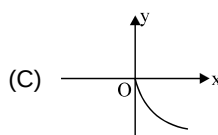
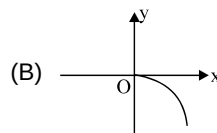
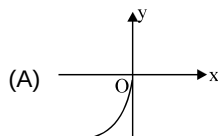
Candidate's Signature _____ Invigilator's Signature: _____

SECTION – I: PHYSICS

PART – A

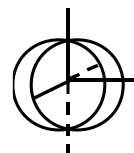
This Section contains **20 Single choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

1. In a region of space uniform electric field is present as $\vec{E} = E_0 \hat{j}$ and uniform magnetic field is present as $\vec{B} = B_0 \hat{k}$. An electron is released from rest at origin. Which of the following best represents the path followed by electron after release.



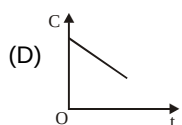
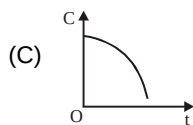
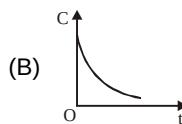
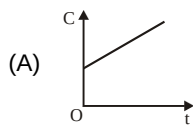
2. Which of the following particle will have minimum frequency of revolution when projected with the same velocity perpendicular to a magnetic field?
- (A) electron (B) proton
(C) He^+ (D) Li^+

3. Two insulated rings, one of slightly smaller diameter than the other, are suspended along their common diameter as shown. Initially the planes of the rings are mutually perpendicular. When a steady current is set up in each of them
- (A) the two rings rotate into a common plane
(B) the inner ring oscillates about its initial position
(C) the outer ring stays stationary while the inner one moves into the plane of the outer ring
(D) the inner ring stays stationary while the outer one moves into the plane of the inner ring.

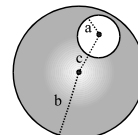


Space for rough work

4. A parallel plate capacitor has a dielectric slab in it. The slab just fills the space inside the capacitor. The capacitor is charged by a battery and then battery is disconnected. Now the slab is started to pull out slowly at $t = 0$. If at time t , capacitance of the capacitor is C , then which of the following graphs is correct?



5. A long, straight metal has a long hole of radius a drilled parallel to the rod axis with cross-sectional view as shown. If the rod carries a current I , the value of the magnetic field on the axis of the rod is-



(A) $B = \frac{\mu_0 I a^2}{2\pi c(b^2 - a^2)}$

(B) $B = \frac{\mu_0 I}{2\pi c}$

(C) $B = \frac{\mu_0 I a^2}{4\pi c(b^2 - a^2)}$

(D) Zero

6. A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E} = E_0 \hat{i}$ and magnetic field $\vec{B} = -B_0 \hat{k}$. Its velocity at $(x, 3, 0)$ is $(4\hat{i} + 3\hat{j})$. The value of x is

(A) $\frac{36E_0 B_0}{qm}$

(B) $\frac{25m}{2qE_0}$

(C) $\frac{10m}{qE_0}$

(D) $\frac{25E_0 B_0}{m}$

7. A conducting rod of mass m and length l is placed in the east-west direction on a frictionless horizontal table. The horizontal component of earth's magnetic field is B from south to north. If a charge q is suddenly passed through the conductor, then the instantaneous velocity acquired by the rod is

(A) Zero

(B) $\frac{B\ell q}{2m}$

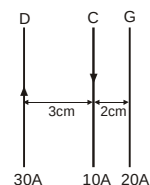
(C) $\frac{B\ell q}{m}$

(D) $\frac{(B\ell q)^2}{m^2}$

Space for rough work

8. Three long straight parallel wires, carrying current, are arranged as shown in figure. The force experienced by a 25 cm length of wire C is

- (A) 10^{-3} N (B) 2.5×10^{-3} N
(C) zero (D) 1.5×10^{-3} N



9. Two long conducting wires are parallel to each other one of them carries a steady electric current I while the other carries an alternating current $I_0 \sin \omega t$. The force between the two wires

- (A) is repulsive (B) is attractive
(C) has average value zero (D) has average value zero if $I = I_0$

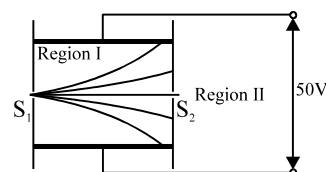
10. Shown in the figure is a conductor carrying a current I . The magnetic field intensity at the point O (common centre of all the three arcs) is

- (A) $\frac{5\mu_0 I \theta}{24\pi r}$ (B) $\frac{\mu_0 I \theta}{24\pi r}$
(C) $\frac{11\mu_0 I \theta}{24\pi r}$ (D) Zero



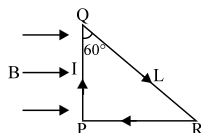
11. At the plates of velocity filter a E.M.F. of 50 V is connected. Distance between the plates is 1/2 metres. We want the particles emerging from slit S_2 in region II to have fixed speed of 1000 m/s. The magnitude of magnetic field applied in region I is

- (A) 0.1 T (B) 0.2 T
(C) 0.5 T (D) 0.25 T



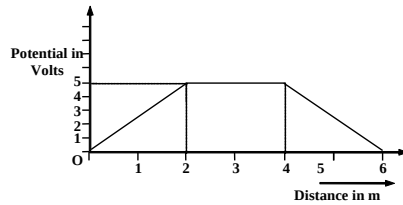
12. For the circuit shown in figure, the direction and magnitude of the force on PQR is:

- (A) Zero (B) ILB out of the page
(C) $ILB/2$ into the page (D) ILB into the page



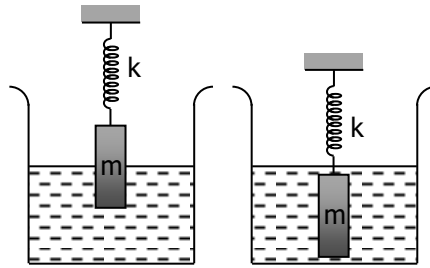
Space for rough work

13. The variation of potential with distance R from fixed point is shown in the figure. The electric field at $R = 5$ m is



- (A) 2.5 V/m
(B) -2.5 V/m
(C) (2/5) V/m
(D) (-2/5) V/m

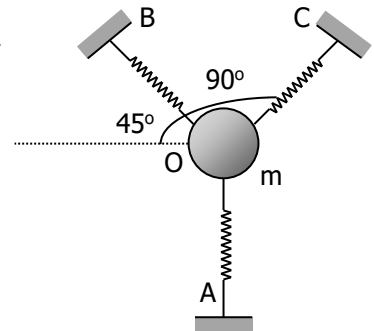
14. Period of small oscillations in the two cases shown in figure is T_1 and T_2 respectively, then:



- (A) $T_1 = T_2$
(B) $T_1 < T_2$
(C) $T_1 > T_2$
(D) cannot say anything

15. A particle of mass m is attached to three identical springs A, B and C each of force constant k as shown in figure. If the particle of mass m is pushed slightly against the spring A and released, then the time period of oscillation is:

- (A) $2\pi\sqrt{\frac{2m}{k}}$
(B) $2\pi\sqrt{\frac{m}{2k}}$
(C) $2\pi\sqrt{\frac{m}{k}}$
(D) $2\pi\sqrt{\frac{m}{3k}}$



Space for rough work

16. The potential energy of a particle of mass 2 kg in SHM is $(9x^2)$ J. Here x is the displacement from mean position. If total mechanical energy of the particle is 36 J. Then maximum speed of the particle is:
 (A) 4 m/s (B) 2 m/s
 (C) 6 m/s (D) 10 m/s

17. A particle is executing vertical SHM about the highest point of a projectile. When the particle is at the mean position, the projectile is fired from the ground with velocity u at an angle θ with the horizontal. The projectile hits the oscillation particle. Then, the time period of the particle cannot be:

- (A) $\frac{u \sin \theta}{g}$ (B) $\frac{2u \sin \theta}{g}$
 (C) $\frac{2u \sin \theta}{3g}$ (D) $\frac{3u \sin \theta}{2g}$

18. The phase difference between the two simple harmonic oscillations

$$y_1 = \frac{1}{2} \sin(\omega t) + \frac{\sqrt{3}}{2} \cos(\omega t)$$

and $y_2 = \sin(\omega t) + \cos(\omega t)$ is:

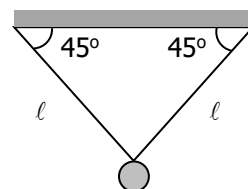
- (A) $\frac{\pi}{2}$ (B) $-\frac{\pi}{6}$
 (C) $\frac{\pi}{12}$ (D) $\frac{7\pi}{12}$

19. What is the ratio between the height H ($\ll$ radius of earth) of a mountain and the depth h of a mine, if a pendulum swings with the same period at the top of the mountain and at the bottom of the mine?

- (A) $1/2$ (B) 2
 (C) 1 (D) 4

20. Time period of small oscillations (in a vertical plane normal to the plane of strings) of the bob in the arrangement shown will be:

- (A) $2\pi\sqrt{\frac{\ell}{g}}$ (B) $2\pi\sqrt{\frac{\ell}{\sqrt{2}g}}$
 (C) $2\pi\sqrt{\frac{\sqrt{2}\ell}{g}}$ (D) $2\pi\sqrt{\frac{2\ell}{g}}$



Space for rough work

PART B**Integer Based**

This section contains 5 questions. The answer to each of the questions is a single digit integer, ranging from 0 to 9. The appropriate bubbles below the respective questions numbers in the ORS have to be darkened. For example, if the correct answers to questions numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:

X	Y	Z	W
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

- A particle of specific charge $q/m = \pi$ C/kg is projected from the origin towards positive x-axis with a velocity of 10 m/s in a uniform magnetic field $\vec{B} = -2\hat{k}$ Tesla. The magnitude of velocity V of the particle after time $t = \frac{1}{6}$ s is $10n$. Then what will be value of n .
- An electron moving with a velocity $\vec{V}_1 = 1\hat{i}$ m/s at a point in a magnetic field experiences a force $\vec{F}_1 = -2\hat{j}$ N. If the electron moving with a velocity $\vec{V}_2 = 1\hat{j}$ m/s at the same point, it experiences a force $\vec{F}_2 = 2\hat{i}$ N. The force that an electron would experience if it were moving with a velocity $\vec{V}_3 = 2\hat{k}$ m/s at the same point is
- Three concentric conducting spherical shells carry charges as follows: $+Q$ on the inner shell, $-2Q$ on the middle shell and $-5Q$ on the outer shell. The charge on the inner surface of the outer shell is $\pm nQ$ then value of n is
- Two pendulums have time periods T and $\frac{5T}{4}$. They start SHM at the same time from the mean position. After how many oscillations of the smaller pendulum they will be again in the same phase:
- The period of particle executing SHM is 8 s. At $t = 0$ it is at the mean position. The ratio of the distances covered by the particle in the 1st to second to the 2nd second is $\sqrt{x} + 1 : 1$ then the value of x is

Space for rough work

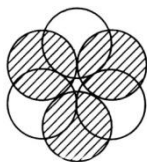
SECTION – II: CHEMISTRY**PART – A**

This Section contains **20 Single choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

- When heated above 916°C , iron changes its bcc crystalline form to fcc without the change in the radius of atom. The ratio of density of the crystal before heating and after heating is:
(A) 1.069 (B) 0.918
(C) 0.725 (D) 1.231
- The density of solid argon ($A_r = 40 \text{ g/mol}$) is 1.68 g/mL at 40 K . If the argon atom is assumed to be a sphere of radius $1.50 \times 10^{-8} \text{ cm}$, what % of solid Ar is apparently empty space? (use $N_A = 6 \times 10^{23}$)
(A) 35.64 (B) 64.36
(C) 74% (D) none of these
- First three nearest neighbor distances for primitive cubic lattice are respectively (edge length of unit cell = a):
(A) $a, \sqrt{2}a, \sqrt{3}a$ (B) $\sqrt{3}a, \sqrt{2}a, a$
(C) $a, \sqrt{2}a, 2a$ (D) $a, \sqrt{3}a, 2a$
- The density of a pure substance 'A' whose atoms pack in cubic close pack arrangement is 1 g/cc . If B atoms can occupy tetrahedral void and if all the tetrahedral voids are occupied by 'B' atom. What is the density of resulting solid in g/cc. [Atomic mass (A) = 30 g/mol and atomic mass (B) = 50 g/mol]
(A) 3.33 (B) 4.33
(C) 2.33 (D) 5.33
- A crystal is made of particles X and Y. X forms fcc packing and Y occupies all the octahedral voids. If all the particles along one body diagonal are removed, then the formula of the crystal would be:
(A) X_4Y_3 (B) X_5Y_4
(C) X_4Y_5 (D) none of these
- An element crystallizes in a structure have fcc unit cell of an edge 200 pm . Calculate the density, if 100 g of this element contains 12×10^{23} atoms:
(A) 41.66 g/cm^3 (B) 4.166 g/cm^3
(C) 10.25 g/cm^3 (D) 1.025 g/cm^3

Space for rough work

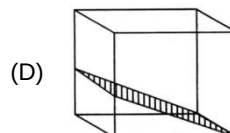
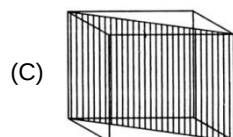
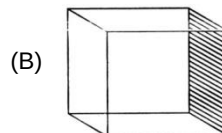
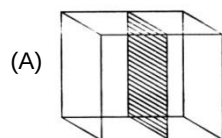
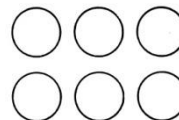
7. The empty space in the following packing is called:



(A) tetrahedral void
(C) triangular void

(B) octahedral void
(D) hexagonal void

8. Which of the following shaded plane in fcc lattice contains arrangement of atoms as shown below?



9. CsBr has bcc structure with edge length $4.3\text{\AA}$. The shortest interionic distance in between Cs^+ and Br^- is
 (A) $3.72\text{\AA}$ (B) $1.86\text{\AA}$
 (C) $7.44\text{\AA}$ (D) $4.3\text{\AA}$
10. The coordination number of a metal crystallizing in a hexagonal close packed structure is:
 (A) 12 (B) 4
 (C) 8 (D) 6
11. In which of the following crystals alternate tetrahedral voids are occupied?
 (A) NaCl (B) ZnS
 (C) CaF_2 (D) Na_2O
12. If $r(\text{Na}^+)/r(\text{Cl}^-) = 0.5$ and $r(\text{Na}^+)/r(\text{K}^+) = 0.7$, then the ratio $a(\text{KCl})/a(\text{NaCl})$ of edge lengths of unit cell is about
 (A) 1.14 (B) 1.24
 (C) 1.34 (D) 1.44

Space for rough work

13. The composition of a sample of wurtzite is $\text{Fe}_{0.93}\text{O}_{1.00}$. The percentage of Fe(III) ions present in the sample is about
 (A) 10% (B) 15%
 (C) 20% (D) 25%
14. The vapour pressure of a liquid in a closed container depends upon (i) temperature of liquid (ii) quantity of liquid, (iii) surface area of liquid:
 (A) (i) only (B) (ii) only
 (C) (i) and (iii) only (D) (i), (ii) and (iii)
15. A mixture of ethyl alcohol and propyl alcohol has a vapour pressure of 290 mm at 300 K. The vapour pressure of propyl alcohol is 200 mm. If the mole fraction of ethyl alcohol is 0.6, its vapour pressure (in mm) at the same temperature will be
 (A) 360 (B) 350
 (C) 300 (D) 700
16. A sample of air is saturated with benzene (vapor pressure of pure benzene = 100 mm Hg at 298 K) at 298 K, 750 mm Hg pressure. If it is isothermally compressed to one third of its initial volume, the final pressure of the system is
 (A) 2250 to rr (B) 2150 to rr
 (C) 2050 to rr (D) 1950 to rr
17. At 40°C the vapour pressure in torr, of methyl alcohol-ethyl alcohol solutions is represented by the equation.
 $P = 119X_A + 135$ where X_A is mole-fraction of methyl alcohol, then the value of $\lim_{X_A \rightarrow 1} \frac{P_A}{X_A}$ is
 (A) 254 torr (B) 135 torr
 (C) 119 torr (D) 140 torr
18. At a certain temperature pure liquid A and liquid B have vapour pressures 10 torr and 37 torr respectively. For a certain ideal solution of A and B, the vapour in equilibrium with the liquid has the components A and B in the partial pressure ratio $P_A : P_B = 1:7$. What is the mole fraction of A in the solution?
 (A) 0.346 (B) 0.654
 (C) 0.5 (D) None of these
19. Two liquids A and B are mixed at temperature T in a certain ratio to form an ideal solution. It is found that the partial vapour pressure of A i.e., P_A is equal to P_B , the vapour pressure of B for the liquid mixture. What is the total vapour pressure of the liquid mixture in terms of P_A° and P_B° ?
 (A) $\frac{P_A^\circ P_B^\circ}{P_A^\circ + P_B^\circ}$ (B) $\frac{2P_A^\circ P_B^\circ}{P_A^\circ + P_B^\circ}$
 (C) $\frac{P_A^\circ}{P_A^\circ + P_B^\circ}$ (D) $\frac{2P_B^\circ}{P_A^\circ + P_B^\circ}$
20. The vapour pressures of ethanol and methanol are 42.0 mm and 88.5 mm Hg respectively. An ideal solution is formed at the same temperature by mixing 46.0 g of ethanol with 16.0 g of methanol. The mole fraction of methanol in the vapour is:
 (A) 0.467 (B) 0.502
 (C) 0.513 (D) 0.556

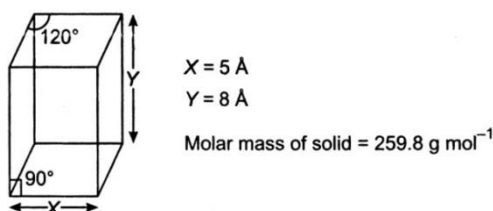
Space for rough work

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2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

1. A solid crystallizes in hexagonal lattice as shown in above figure. Density of the solid is 5 g/cm^3 . How many molecules are there in the given unit cell?



(Avogadro's number = 6.023×10^{23}):

2. 3L of moist gas at 3 atm pressure is completely transferred to a 1L container at the same temperature. If the vapour pressure of water is 0.5 atm at that temperature, what is the pressure of moist gas in the 1L container?
3. Atoms 'A' form F.C.C., atoms 'B' occupy half of octahedral voids and atoms 'C' occupy all the tetrahedral voids. The empirical formula of the compound is $A_xB_yC_z$. Calculate $(x + y + z)$
4. If the co-ordination number of atoms in h, c, p is 'x' and that of simple cubic system is 'y', calculate $\left(\frac{x}{y}\right)$.
5. The vapour pressures of 'A' and 'B' in the pure state are 600 mmHg and 800 mmHg respectively. The mole traction of 'A' in the binary liquid solution is 0.25. If the mol% of 'A' in vapour phase is x%, report the value of $\left(\frac{x}{5}\right)$.

Space for rough work

SECTION – III: MATHEMATICS

PART – A

This Section contains **20 Single choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

- $\int \frac{\sin x}{\cos^2 x \cdot \sqrt{\cos 2x}} dx$ equals:
 (A) $C - \sqrt{1 - \tan^2 x}$ (B) $C - \sqrt{1 + \sec^2 x}$
 (C) $C - \sqrt{1 + \cos^2 x}$ (D) $C - \sqrt{1 - \sin^2 x}$
- $\int x^x \left(\frac{1}{x} + \ln^2 x + \ln x \right) dx$ is equal to (where C is constant of integration):
 (A) $x^x \left(\ln^2 x - \frac{1}{x} \right) + C$ (B) $x^2 (\ln x - x)$
 (C) $\frac{x^x \ln^2 x}{2} + C$ (D) $x^x \ln x + C$
- If $\int e^x \left(\frac{x+2}{x+4} \right)^2 dx = \frac{f(x)e^x}{x+4} + C$ then the sum $f(x) + f^2(x) + f^3(x) \dots \infty$ at $x = \frac{1}{2}$, is
 (A) $\frac{1}{2}$ (B) $\frac{1}{4}$
 (C) 1 (D) $\frac{3}{2}$
- $\int \frac{(2x+1)}{(x^2+4x+1)^{3/2}} dx$ equals
 (A) $\frac{x^3}{(x^2+4x+1)^{1/2}} + C$ (B) $\frac{x}{(x^2+4x+1)^{1/2}} + C$
 (C) $\frac{x^2}{(x^2+4x+1)^{1/2}} + C$ (D) $\frac{1}{(x^2+4x+1)^{1/2}} + C$
- The evaluation of $\int \frac{px^{p+2q-1} - px^{q-1}}{x^{2p+2q} + 2x^{p+q} + 1} dx$ is
 (A) $-\frac{x^p}{x^{p+q} + 1} + C$ (B) $\frac{x^q}{x^{p+q} + 1} + C$
 (C) $-\frac{x^q}{x^{p+q} + 1} + C$ (D) $\frac{x^p}{x^{p+q} + 1} + C$

Space for rough work

6. If $f(x) = \int 2e^x \cos^2 x (-\tan^2 x + \tan x + 1) dx$ and $f(x)$ passes through $(\pi, 0)$, then $(f(0) + f'(0))$ equals:

(A) $2e$ (B) $3e$
(C) 2 (D) 3

7. $\int \left(\frac{g'(x)}{g(x)} - \frac{f'(x)}{f(x)} \right) (\ln(g(x)) - \ln(f(x))) dx$ equal to

(A) $\ln \left(\frac{g(x)}{f(x)} \right) + C$ (B) $\frac{1}{2} \left(\ln \left(\frac{g(x)}{f(x)} \right) \right)^2 + C$
(C) $\frac{g(x)}{f(x)} \ln \left(\frac{g(x)}{f(x)} \right) + C$ (D) $\ln \left(\frac{f(x)}{g(x)} \right) + C$

8. $\int e^{\sin x} \left(\frac{x \operatorname{cosec} x - \sec^3 x}{\operatorname{cosec} x \cdot \sec x} \right) dx$ equals

(A) $e^{\sin x} - e^{\sin x} \sec x + C$ (B) $ex^{\sin x} + e^{\sin x} \sec x + C$
(C) $xe^{\sin x} - e^{\sin x} \sec x + C$ (D) $xe^{\sin x} - \sec x + C$

9. $\int \frac{dx}{\sqrt{x} - \sqrt[3]{x}}$ equals

(A) $2\sqrt{x} + 3\sqrt[3]{x} + 6\sqrt[6]{x} + 6 \ln |\sqrt[6]{x} - 1| + C$ (B) $2\sqrt{x} + 3\sqrt[3]{x} - 6\sqrt[6]{x} + 6 \ln |\sqrt[6]{x} - 1| + C$
(C) $3\sqrt{x} + 2\sqrt[3]{x} + 6\sqrt[6]{x} + 6 \ln |\sqrt[6]{x} - 1| + C$ (D) $3\sqrt{x} + 2\sqrt[3]{x} + 6\sqrt[6]{x} - \ln |\sqrt[6]{x} - 1| + C$

10. If $\int x^2 e^{-2x} dx = e^{-2x} (ax^2 + bx + c) + d$, then:

(A) $a = -\frac{1}{2}, b = -\frac{1}{2}, c = -\frac{1}{4}$ (B) $a = -\frac{1}{2}, b = -\frac{1}{2}, c = \frac{1}{4}$
(C) $a = -\frac{1}{2}, b = -1, c = -\frac{1}{2}$ (D) $a = 1, b = 1, c = -\frac{1}{2}$

Space for rough work

11. $\int \sec^2 \theta (\sec \theta + \tan \theta)^2 d\theta$
- (A) $\frac{(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$ (B) $\frac{(\sec \theta + \tan \theta)}{3} [2 + 4 \tan \theta (\sec \theta + \tan \theta)] + C$
- (C) $\frac{(\sec \theta + \tan \theta)}{3} [1 + \sec \theta (\sec \theta + \tan \theta)] + C$ (D) $\frac{3(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$
12. $\int e^x \frac{(1+n \cdot x^{n-1} - x^{2n})}{(1-x^n)\sqrt{1-x^{2n}}} dx$ is equal to
- (A) $e^x \sqrt{\frac{1-x^n}{1+x^n}} + C$ (B) $e^x \sqrt{\frac{1+x^n}{1-x^n}} + C$
- (C) $-e^x \sqrt{\frac{1-x^n}{1+x^n}} + C$ (D) $-e^x \sqrt{\frac{1+x^n}{1-x^n}} + C$
13. The value of the integral $\int \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x} dx$ is:
- (A) $\sin x - 6 \tan^{-1}(\sin x) + c$ (B) $\sin x - 2 (\sin x)^{-1} + c$
- (C) $\sin x - 2 (\sin x)^{-1} - 6 \tan^{-1}(\sin x) + c$ (D) $\sin x - 2 (\sin x)^{-1} + 5 \tan^{-1}(\sin x) + c$
14. If $\int \frac{dx}{x\sqrt{1-x^{2003}}} = k \cdot \log \left| \frac{\sqrt{1-x^{2013}} - 1}{\sqrt{1-x^{2013}} + 1} \right| + C$, then the value of $\left(\frac{1}{k}\right)$ is equal to (where C is constant of integration)
- (A) 2010 (B) 2011
- (C) 2012 (D) 2013
15. If $\int \frac{x^3 - 8}{x^3 \sqrt{x^4 - x^3 + 2}} = dx = \frac{k \cdot \sqrt{x^4 - x^3 + 2}}{x^2} + C$, then the value of k is (where C is constant of integration):
- (A) 3 (B) 2
- (C) 7 (D) 4
16. $\int \frac{\cos^6 x \cdot dx}{\sin^4 x (\cos^7 x + \sin^7 x)^{\frac{4}{7}}}$ is equal to:
- (A) $\frac{-1}{3} (1 + \cot^7 x)^{\frac{3}{7}} + C$ (B) $\frac{-1}{3} (1 + \cot^7 x)^{\frac{7}{3}} + C$
- (C) $\frac{-1}{21} (1 + \cot^7 x)^{\frac{3}{7}} + C$ (D) $\frac{1}{21} (1 + \cot^7 x)^{\frac{7}{3}} + C$

Space for rough work

17. If $6^{83} + 8^{83}$ is divided by 49, then the remainder is
(A) 35 (B) 5
(C) 1 (D) 0
18. Let $\binom{n}{k}$ represents the combination of 'n' things taken 'k' at a time, then the value of the value of sum
 $\binom{99}{97} + \binom{98}{96} + \binom{97}{95} + \dots + \binom{3}{1} + \binom{2}{0}$ equals
(A) $\binom{99}{97}$ (B) $\binom{100}{98}$
(C) $\binom{99}{98}$ (D) $\binom{100}{97}$
19. The value of expression ${}^{n+1}C_2 + 3({}^2C_2 + {}^3C_2 + {}^4C_2 + \dots + {}^nC_2)$ is.
(A) $\sum n^3$ (B) $\sum n^2$
(C) $\sum n$ (D) $\frac{n^2(n+1)}{2}$
20. The value of $\sum_{r=0}^{50} (-1)^r \frac{{}^{50}C_r}{(r+1)}$ is
(A) $\frac{1}{50}$ (B) $\frac{50}{51}$
(C) $\frac{1}{51}$ (D) $\frac{51}{50}$

Space for rough work

PART B**Integer Based**

This section contains 5 questions. The answer to each of the questions is a single digit integer, ranging from 0 to 9. The appropriate bubbles below the respective questions numbers in the ORS have to be darkened. For example, if the correct answers to questions numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:

X	Y	Z	W
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

- Let $f(x) = \int (x \cos x + 1)e^{\sin x} dx$ and $g(x) = \int (x \sin x - 1)e^{\cos x} dx$. Further suppose that $f(0) = g(0) = 0$. Find the number of distinct solution (s) of the equation $f(x) \cdot g(x) = 0$.
- $\int \frac{(1+x^{1/4})^{1/3}}{\sqrt{x}} dx = \frac{a}{b} (1+x^{1/4})^{7/3} - c(1+x^{1/4})^{4/3} + k$; a and b are coprime. Find value of $(a-b-c)$
- If $\int \frac{(3x^2 + 2x)}{(x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5)} dx - A \ln \left(\frac{g(x)}{B} \right) + c$, where $g(x)$ is a polynomial function with leading coefficient 1 and c is integration constant, then find the value of B/A .
- Find the value of $\sum_{k=0}^{21} k \cdot {}^{21}C_k \left(\frac{1}{3} \right)^k \left(\frac{2}{3} \right)^{21-k}$
- If the coefficient of first three terms in the expansion $\left(\sqrt{x} + \frac{1}{2\sqrt[4]{x}} \right)^n$ where $n \in \mathbb{N}$ form an arithmetic progression, then find the number of terms in the expansion having integral powers of x.

Space for rough work
